Divya Ramesh

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EDUCATION

Columbia University New York, NY
M.S. in Computer Science (Natural Language Processing) *Dec 2026*
- *Selected Coursework:* Natural Language Generation & Summarization, Machine Learning, Natural Language Processing, Spoken Language Processing, Designing LLM Agents, Neural Networks & Deep Learning

University of California, Los Angeles Los Angeles, CA
B.A. in Psychology, B.A. in English (GPA: 3.94 / 4.0) *Jun 2024*
- *Selected Coursework:* Neural Networks, Cognitive Psychology, Social Psychology, Statistics
- *Honors:* Cum Laude, Dean's Honor List (all quarters), Phi Beta Kappa

SKILLS

Languages: Python, C/C++, R, Java, JavaScript, SQL
ML / LLM Frameworks & Tools: PyTorch, Databricks, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-Learn, NumPy, XGBoost
Domains & Techniques: Agentic AI Systems, LLM-as-a-Judge Evaluation, LLM Fine-Tuning, Preference Learning (Pairwise Ranking), Supervised Learning, Reinforcement Learning Concepts, Feature Engineering, RAG Systems

EXPERIENCE

Comcast New York, NY
AI Research Intern *June 2026 - August 2026*
- Designed and deployed LLM-as-a-Judge evaluation frameworks on Databricks to systematically score trajectory outputs, response quality, and performance of production multi-turn agentic AI systems.
- Architected a pairwise learning-to-rank pipeline utilizing production execution logs and human annotations to train and optimize a RAG-based conversational agent.
- Engineered features and implemented a supervised learning re-ranking strategy for document retrieval, successfully balancing broad query relevance with support for targeted, niche search requests.
- Measurably boosted system accuracy, improving overall retrieval quality by 4% and final answer quality by 2%.

Spoken Language Processing Group New York, NY
Research Assistant *Sep 2025 - Present*
- Lead research on evaluating Large Language Models (LLMs) for clinical NLP tasks, measuring cultural competency, alignment, and performance across complex dialogue datasets.
- Build robust LLM evaluation and benchmarking frameworks to drive model selection, bias assessment, and guardrailing for deployment in therapeutic conversational AI systems.

PROJECTS

Investigating Poetic Device Constraints on LLM Chain-of-Thought (CoT) Reasoning New York, NY
Chain-of-Thought (CoT), Qwen, Llama 3.1, LLM-as-a-Judge, GCP *Oct 2026 - Dec 2026*
- Designed and executed a controlled ablation study evaluating the impact of prosodic/poetic formatting constraints (e.g., rhyme schemes, syllable counts, alliteration) on open-weight LLMs (Qwen2.5-7B, Llama-3.1-8B) across mathematical (GSM8K) and logical (LogiQA v2) reasoning tasks.
- Developed an LLM-as-a-judge evaluation framework using custom scoring rubrics to quantify constraint adherence, logic/calculation errors, and text quality (ROUGE-L F1), validated via inter-annotator agreement (weighted Kappa).
- Configured and deployed quantized model inference on Google Cloud GPUs to efficiently run cross-domain experiment benchmarks under compute constraints.

Decentralized P2P Payment Application New York, NY
Python, Blockchain, Socket Programming, Cryptography, Distributed Systems *April 2026 - May 2026*
- Engineered a decentralized platform featuring custom proof-of-work consensus, multi-threaded peer node handling, transaction validation, and fork resolution algorithms for reliable distributed ledger state.

Speech Emotion Recognition Classifier New York, NY
scikit-learn, openSMILE, Parselmouth *April 2026*
- Developed an end-to-end ML pipeline extracting 384 acoustic features across 15 emotion categories with per-speaker normalization, achieving ~4x above-chance classification accuracy using Leave-One-Group-Out cross-validation.